

| Direct Qwen3 8B |          |                 |              | Final selected pipeline |          |                 |              |             |
|-----------------|----------|-----------------|--------------|-------------------------|----------|-----------------|--------------|-------------|
| Gold label      |          | Negative        | Neutral      | Positive                |          | Negative        | Neutral      | Positive    |
|                 | Negative | 66<br>80.5%     | 13<br>15.9%  | 3<br>3.7%               | Negative | 70<br>85.4%     | 9<br>11.0%   | 3<br>3.7%   |
|                 | Neutral  | 59<br>23.8%     | 146<br>58.9% | 43<br>17.3%             | Neutral  | 50<br>20.2%     | 160<br>64.5% | 38<br>15.3% |
|                 | Positive | 6<br>5.4%       | 18<br>16.1%  | 88<br>78.6%             | Positive | 3<br>2.7%       | 19<br>17.0%  | 90<br>80.4% |
|                 |          | Predicted label |              |                         |          | Predicted label |              |             |

Row-normalized: each cell shows count and row percentage